



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2025 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

4045/01

Paper 1

31 July 2025

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** questions.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142, unless the question requires the answer in terms of π .

Total Marks

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f} \right)^2}$$

Answer **all** the questions.

- 1 (a) Write 0.0068579 correct to 3 significant figures.

Answer [1]

- (b) Find 125.5% of \$320.

Answer \$ [1]

-
- 2 $(9.8 \times 10^a) \times (2.1 \times 10^b) = m \times 10^n$ where $1 \leq m < 10$ and a, b and n are integers.

- (a) Find the value of m .

Answer $m =$ [1]

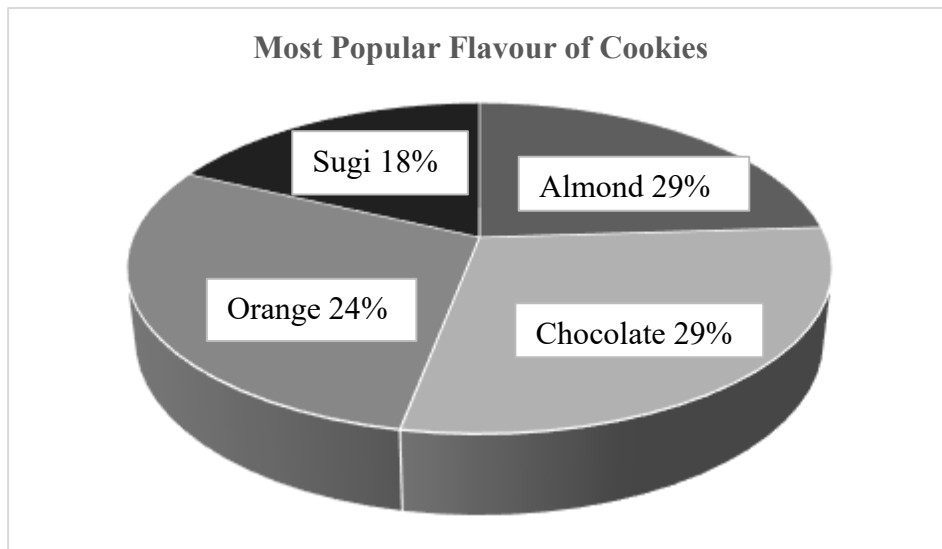
- (b) Write an expression for n in terms of a and b .

Answer $n =$ [1]

-
- 3 Simplify $3(a + 4b) - 5b - 4a$.

Answer [2]

- 4 The pie chart shows the sales for four different flavours of cookies.



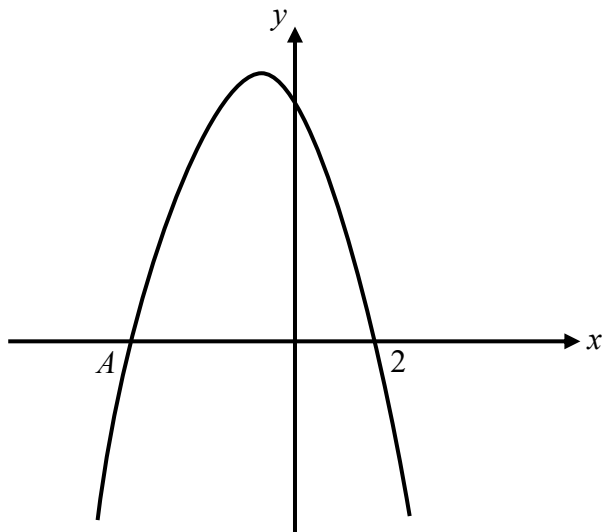
- (a) State one aspect of the graph that may be misleading.

.....
..... [1]

- (b) Explain how this may lead to a misinterpretation of the graph.

.....
..... [1]

- 5 The graph of the quadratic function $y = -(x+4)(x-2)$ is shown below.



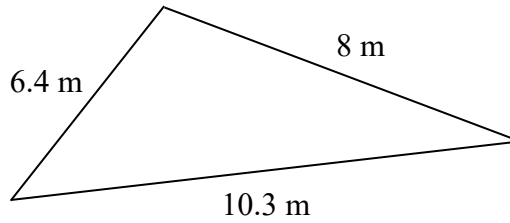
- (a) Write down the coordinates of A .

Answer A (..... ,) [1]

- (b) Write down the equation of the line of symmetry.

Answer [1]

- 6 Sam is planning to build a triangular path to go around his garden. The dimensions of the triangle are shown in the diagram below.



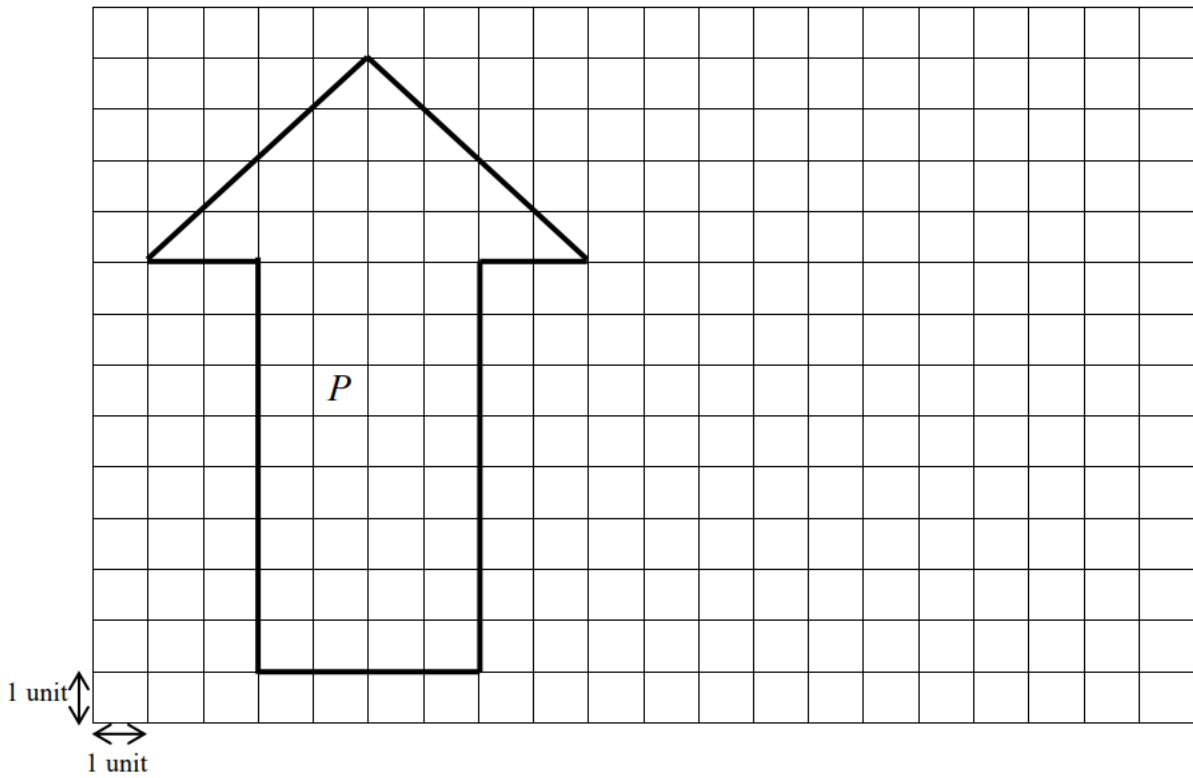
The path must form a right-angled triangle so that it fits nicely around the garden.

Based on the given dimensions, show your calculations clearly to determine whether the path will fit nicely around the garden.

Answer

[2]

- 7 On the grid, draw a reduction of Figure P using a scale factor of $\frac{1}{2}$.



[2]

- 8 The stem-and-leaf diagram shows the prices of books (in dollars) at a bookstore.

1	0	2	5	8
2	1	3	7	
3	0	5		
4	2			
5				
6				
7	9			

Key:

1 | 0 means \$10

Find the

- (a) mean price,

Answer \$ [1]

- (b) median price.

Answer \$ [1]

- (c) Which average, mean or median, is more suitable for summarising the prices of books at the bookstore?

Give a reason for your answer.

.....

..... [1]

- 9 Factorise $6ax - 3ay + 14bx - 7by$ completely.

Answer..... [2]

- 10 A runner jogs at 9 km/h for 45 minutes.

- (a) Convert 9 km/h to m/s.

Answer m/s [1]

- (b) Find the distance covered by the runner, giving your answer in metres.

Answer m [1]

11 (a) Solve $5 - \frac{7}{4}x = 2$.

Answer $x = \dots\dots\dots$ [1]

- (b) y is directly proportional to x^2 .
When $x = \frac{4}{3}$, $y = 3.2$.
Find the value of y when $x = 1.5$.

Answer $y = \dots\dots\dots$ [2]

- 12 (a) Write $\left(\frac{4b^2}{9a^2}\right)\left(\frac{3a^3}{2b^3}\right)$ as a single fraction in its simplest terms.

Answer [2]

- (b) Find the largest integer satisfying $5x < 13$.

Answer [2]

-
- 13 Solve these simultaneous equations.

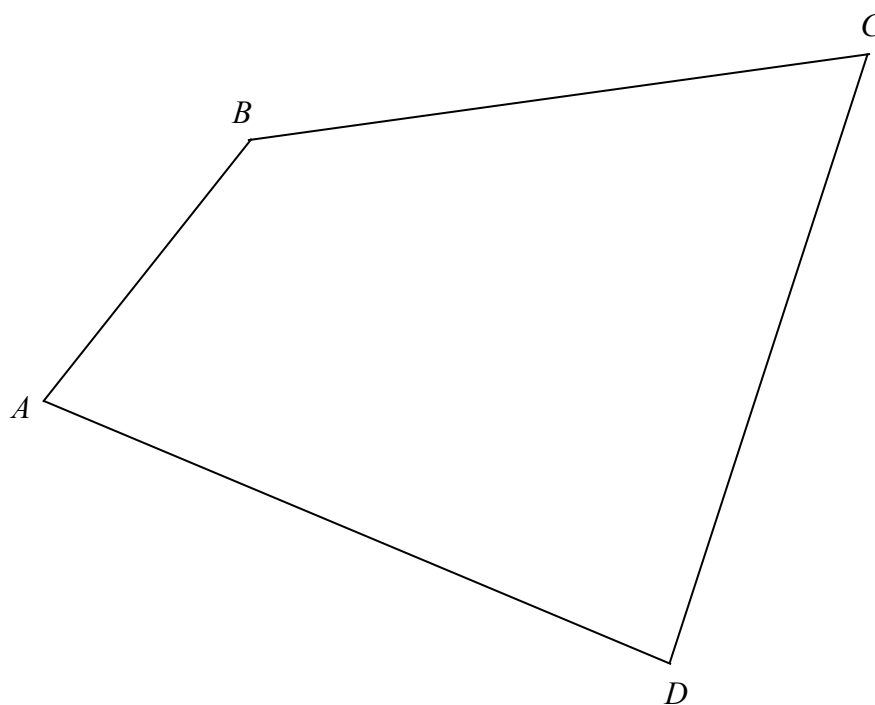
$$2x = 9 - y$$

$$3x - 2y = 24$$

Show your working.

Answer $x =$

$y =$ [3]



- (a) Construct the angle bisector of angle BAD . [1]
- (b) Construct the perpendicular bisector of AD . [1]
- (b) The angle bisector and the perpendicular bisector meet at X .
Measure DX .

Answer $DX = \dots\dots\dots$ cm [1]

15 $H = 2^2 \times 3^3 \times 7$ and $K = 2^3 \times 3^2 \times 5^2$.

- (a) Find the lowest common multiple of H and K , giving your answer as a product of its prime factors.

Answer [1]

- (b) Find the smallest integer m such that the product of $m \times H \times K$ is a perfect square.

Answer [1]

16 By completing the square, solve $x^2 - 10x + 6 = 0$.

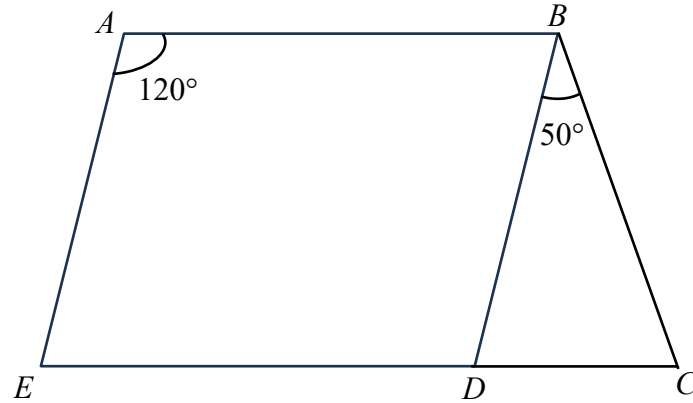
Show your working and give your answers correct to 2 decimal places.

Answer $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

- 17 A parallelogram $ABDE$ and triangle BCD are joined as shown.

Angle $EAB = 120^\circ$ and angle $DBC = 50^\circ$.

EDC is a straight line.



Ryan says that triangle BCD is an isosceles triangle. Is he correct?

Show all your working clearly.

Give a reason for each step of your working.

Answer

..... [3]

- 18 The manager of a gym records the time the members spend exercising in the gym on a particular morning.

The data is presented in the frequency table below.

Time (minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	24
$20 < t \leq 30$	18
$30 < t \leq 40$	15
$40 < t \leq 50$	10
$50 < t \leq 60$	?

The manager forgets how many members exercised for more than 50 minutes, up to and including 60 minutes, i.e. $50 < t \leq 60$.

The probability that a member spent exercising for 30 minutes or less was 0.5.

Use the information to find the number of members in the interval $50 < t \leq 60$.

Answer [3]

- 19 An open container is in the shape of a cylinder.
 The height of the cylinder is twice its diameter.
 Given that the radius is x cm, find an expression, in terms of x , for the **total exterior surface area** of the container.

Answer cm^2 [3]

20 Ella runs a bakery and delivers cakes to customers.

- (a) She delivers 120 cakes and earns \$3600 from the delivery of all the cakes.
Calculate how much she earns per cake delivered.

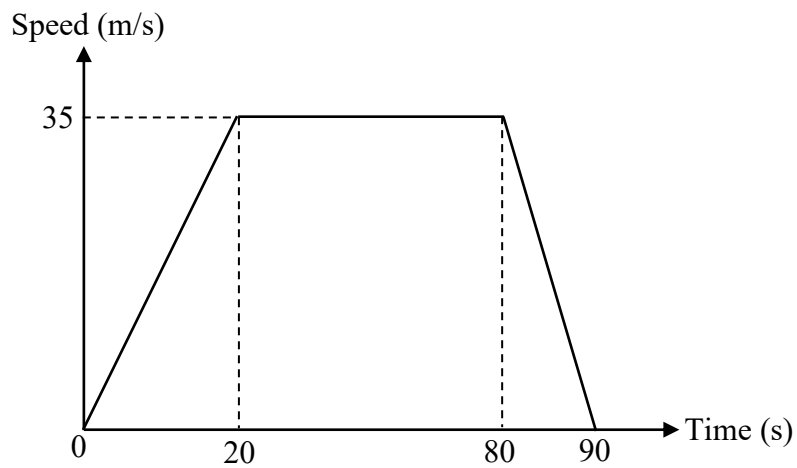
Answer \$ [1]

- (b) Ella needs to deliver 10 cakes on a particular day.
She starts her deliveries at 9:30 a.m.
Each delivery takes 18 minutes, and she takes a 15-minute break after every 4 deliveries.

At what time will she finish all 10 deliveries?

Answer [2]

- 21 The diagram shows the speed-time graph of a journey.



- (a) Find the acceleration for the first 20 seconds of the journey.

Answer m/s^2 [1]

- (b) Calculate the speed at the first 5 seconds of the journey.

Answer m/s [1]

- (c) The area under the graph represents the total distance travelled.

Calculate the total distance travelled.

Answer m [2]

22 An aviation chart is drawn to the scale 1 : 75 000.

- (a) A runway measures 4.8 cm on the chart.
Find the actual length of the runway.
Give your answer in km.

Answer km [2]

- (b) The actual area of an airfield is 1.2 km².
Find the area of the airfield on the chart.
Give your answer in cm².

Answer cm² [2]

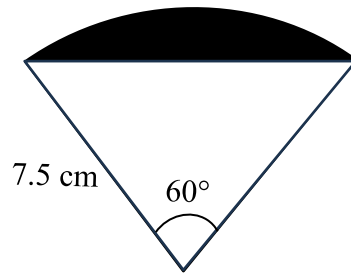
23 Solve $\frac{2}{x-3} - \frac{1}{x+2} = 4$.

Give your answers correct to 2 decimal places.

Answer $x =$ or $x =$ [4]

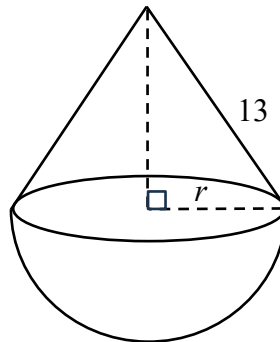
- 24 (a) A sector has radius 7.5 cm and angle 60° .

Find the area of shaded segment.



Answer cm^2 [3]

- (b) A toy is made by attaching a cone to the top of a hemisphere.



Both the cone and the hemisphere have the same radius, r cm.

The slant height of the cone is 13 cm.

The curved surface area of the hemisphere is $50\pi \text{ cm}^2$.

Find the volume of the toy.

Answer cm^3 [4]

END OF PAPER

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